

MH-SIGNALS: Multi-Signal Mental Health Detection with Retrieval-Grounded Support

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Abstract

Online mental health discussions present challenges for timely and consistent moderation due to their sensitive nature (De Choudhury et al., 2013; Coppersmith et al., 2015; Matero et al., 2019). We introduce **MH-SIGNALS**, a framework that combines multi-signal mental health detection with retrieval-grounded support. The system jointly predicts communicative intent and concern severity (low, medium, high).

To overcome limitations of keyword-based approaches, we integrate embedding-based models and transformer architectures (DistilRoBERTa, RoBERTa) (Devlin et al., 2019; Liu et al., 2019) with parameter-efficient fine-tuning (LoRA) (Hu et al., 2021; Houlsby et al., 2019). For high-risk cases, a retrieval-augmented pipeline (Lewis et al., 2020) retrieves counselor-style responses from a curated knowledge base to generate grounded, empathetic, and safety-aware outputs.

We evaluate classification using macro-F1, response quality via BERTScore, and grounding via retrieval recall, along with a manual safety audit. Results show that combining classification with retrieval grounding improves detection accuracy and response reliability.

1 Completed Tasks

MH-SIGNALS detects mental-health signals in online posts via (1) multi-label intent classification and (2) single-label concern-level prediction.

1.1 Data Preparation and Labeling

We collected ~6,000 Reddit posts (2022–2025) and applied preprocessing (case normalization, URL/emoji removal, deduplication).

- **Intent:** Critical Risk, Mental Distress, Maladaptive Coping, Positive Coping, Seeking Help, Progress Update, Mood Tracking, Cause of Distress, Miscellaneous
- **Concern:** Low, Medium, High A gold set of 500 posts was manually annotated. Weak supervision (≥ 0.85 confidence) expanded the dataset to ~8,000 posts. Data was split 70/15/15 with stratification.

1.2 Model Architectures

- **Sentence-Embedding Baseline:** Posts are encoded into 384-d vectors using `all-MiniLM-L6-v2`. A One-vs-Rest logistic regression (balanced weights) handles intent classification, while a multi-class logistic regression predicts concern level. This lightweight model trains in seconds and serves as an interpretable baseline.
- **DistilRoBERTa with LoRA Adapters:** We apply LoRA (Hu et al., 2021) to `distilroberta-base` (rank = 64, α = 128), training only low-rank adapters while freezing the backbone. We use weighted binary cross-entropy (intent) and cross-entropy (concern), with a cosine schedule and warm-up. The best checkpoint is selected using validation PR-AUC (intent) and macro-F1 (concern).
- **RoBERTa-Large with LoRA and Focal Loss.** We fine-tune `roberta-large` with LoRA (rank = 16, α = 32, dropout = 0.1), reducing memory by ~60%. Intent classification uses Focal Loss (γ = 2.0) to emphasize hard examples. Training includes gradient accumulation (effective batch size 4 $\times$) and bf16 precision. We

tune decision thresholds (0.05–0.95) on validation to maximize macro-F1.

1.3 Preliminary Results

Table 1 and Table 2 report macro-F1 on the test set.

Model		Intent (F1)
MiniLM + Logistic Regression		0.58
DistilRoBERTa	+	0.64
LoRA		
RoBERTa-Large	+	0.70
LoRA + Focal		

Table 1: Intent classification performance (macro-F1).

Model		Concern (F1)
MiniLM + Logistic Regression		0.69
DistilRoBERTa	+	0.78
LoRA		
RoBERTa-Large	+	0.76
LoRA + Focal		

Table 2: Concern-level classification performance (macro-F1).

2 Potential Risks and Challenges

2.1 Data Imbalance and Label Ambiguity

The dataset is skewed toward low-concern posts, reducing sensitivity to high-risk cases. Overlapping intent categories (e.g., Mental Distress, Cause of Distress, Maladaptive Coping) further introduce ambiguity in multi-label classification.

2.2 Subtle Concern-Level Distinctions

Distinguishing Medium vs. High concern relies on subtle emotional cues rather than explicit lexical signals, leading to frequent boundary misclassifications.

2.3 Hard and Minority Examples

Rare and complex cases are underrepresented, limiting performance on ambiguous or multi-intent posts.

2.4 Generalization

Models may overfit to Reddit-specific language and struggle to generalize across diverse writing styles and communities.

2.5 Computational Constraints

Large transformer models require significant resources, limiting scalability despite parameter-efficient methods like LoRA.

3 Future Work

3.1 Dataset Refinement

We will expand manual annotations beyond the 500-post subset and revisit uncertain labels to reduce noise and ambiguity in overlapping intent categories.

3.2 Minority and Hard Cases

To improve sensitivity to high-risk posts, we will apply class-aware sampling, targeted augmentation, and threshold tuning, focusing on reducing Medium vs. High confusion.

3.3 Relational Intent Modeling

We will model dependencies between co-occurring intent labels using hierarchical or label-aware approaches to better capture semantic overlap.

3.4 Retrieval-Grounded Pipeline

We plan to build a curated knowledge base of coping strategies and crisis resources, enabling grounded and safety-aware support based on predicted labels.

4 Individual Contributions

Ramya Sri Sushma Vasamsetti: Data collection, preprocessing pipeline (normalization, URL/emoji removal, deduplication), and annotation coordination, including resolving disagreements and analyzing data quality issues.

Paavani Ramakrishna: Annotation schema design (intent + concern), label normalization, stratified dataset splits, and weak supervision to scale labeling.

Harsh Kelawala: Model development (MiniLM, DistilRoBERTa, RoBERTa-Large), LoRA integration (PEFT), loss functions, and threshold tuning for multi-label prediction.

Mansi Suryawanshi: Training experiments, evaluation, error analysis, and identification of key failure modes (e.g., Medium vs. High confusion).

Shruti Dhamecha: Configuration system (YAML), experiment reproducibility, logging, and project infrastructure setup.

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